

Performance Comparison of SVM Kernels on Fashion MNIST Dataset

Aim

To implement and compare SVM classifiers with Linear, Polynomial, and RBF kernels on the Fashion MNIST dataset and analyze their performance.

Theory

Support Vector Machines (SVM) use different **kernel functions** to handle different types of data:

- **Linear Kernel** → Straight-line separation
- **Polynomial Kernel** → Curved boundaries (controlled by degree)
- **RBF Kernel** → Highly flexible, non-linear boundaries

Dataset

We use the Fashion MNIST dataset.

- 70,000 grayscale images (28×28 pixels)
- 10 classes (T-shirt, shoe, bag, etc.)

Algorithm / Procedure

1. Load dataset
2. Normalize pixel values
3. Flatten images into vectors
4. Split into training and testing sets
5. Train SVM with:
 - Linear kernel
 - Polynomial kernel
 - RBF kernel
6. Evaluate accuracy
7. Compare results

Program

```
import numpy as np

from sklearn import svm, datasets

from sklearn.metrics import accuracy_score
```

```
from sklearn.model_selection import train_test_split

# Load dataset

from keras.datasets import fashion_mnist

(X_train, y_train), (X_test, y_test) = fashion_mnist.load_data()

# Reduce size (for faster training in lab)

X_train = X_train[:5000]
y_train = y_train[:5000]
X_test = X_test[:1000]
y_test = y_test[:1000]

# Normalize

X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten images

X_train = X_train.reshape(len(X_train), -1)
X_test = X_test.reshape(len(X_test), -1)

# ----- Linear SVM -----

linear_model = svm.SVC(kernel='linear')
linear_model.fit(X_train, y_train)
y_pred_linear = linear_model.predict(X_test)

# ----- Polynomial SVM -----

poly_model = svm.SVC(kernel='poly', degree=3)
poly_model.fit(X_train, y_train)
```

```
y_pred_poly = poly_model.predict(X_test)
```

```
# ----- RBF SVM -----
```

```
rbf_model = svm.SVC(kernel='rbf')
```

```
rbf_model.fit(X_train, y_train)
```

```
y_pred_rbf = rbf_model.predict(X_test)
```

```
# Accuracy
```

```
print("Linear SVM Accuracy:", accuracy_score(y_test, y_pred_linear))
```

```
print("Polynomial SVM Accuracy:", accuracy_score(y_test, y_pred_poly))
```

```
print("RBF SVM Accuracy:", accuracy_score(y_test, y_pred_rbf))
```

Result

The SVM classifiers with different kernels were successfully implemented and evaluated.

- The **RBF kernel achieved the highest accuracy**, demonstrating its ability to model complex patterns in image data.
- The **Linear kernel performed faster** but with lower accuracy.
- The **Polynomial kernel showed moderate performance**, depending on the degree parameter.